

EC400: Dynamic Systems

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Introduction

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 - analytical solution for models with 2 dynamic variables
 - numerical solution of large-scale models (Dynare)
 - phase diagrams and impulse responses

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- General algorithm:
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- The method
 - works fast and can accommodate large systems
 - works poorly for large shocks and highly non-linear systems

FIRST-ORDER LINEAR EQUATION

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- The steady state is **stable** if $x_t \rightarrow \bar{x}$ as $t \rightarrow \infty$
 - a) if $|\lambda| < 1 \Rightarrow$ the steady state is stable $x_t \rightarrow 0$
 - b) if $|\lambda| > 1 \Rightarrow$ the steady state is unstable $x_t \rightarrow \infty$

LINEAR SYSTEMS

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- special case: **decoupled systems** with diagonal A
- general case: transform system into a decoupled one

Matrix Diagonalization

- **Theorem:** if the eigenvalues λ_1, λ_2 of matrix A are real and different, then

$$A = Q\Lambda Q^{-1}, \quad \Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \text{ and } Q = (\nu^1 \ \nu^2),$$

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- The approach
 - straightforward to generalize for systems with > 2 variables
 - can accommodate complex and repeated eigenvalues

Example

Solve dynamic system

$$x_{t+1} = Ax_t, \quad A = \begin{pmatrix} 1 & 0.5 \\ 1 & 1.5 \end{pmatrix}, \quad x_0 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

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① Find eigenvalues $\lambda_1 = 2$ and $\lambda_2 = 0.5$:

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 0.5 \\ 1 & 1.5 - \lambda \end{vmatrix} = \lambda^2 - 2.5\lambda + 1 = 0$$

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- ② Find eigenvectors and matrix $Q = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$:

$$A - \lambda_1 I = \begin{pmatrix} -1 & 0.5 \\ 1 & -0.5 \end{pmatrix} \Rightarrow v^1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$A - \lambda_2 I = \begin{pmatrix} 0.5 & 0.5 \\ 1 & 1 \end{pmatrix} \Rightarrow v^2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

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- ③ Find matrix $Q^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$ applying the formula

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

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- 4 Solve for x_t :

$$\begin{aligned} x_t &= Q\Lambda^t Q^{-1}x_0 = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2^t & 0 \\ 0 & 0.5^t \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 2^t + 0.5^t \\ 2 \cdot 2^t - 0.5^t \end{pmatrix} \end{aligned}$$

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 - Interpret dynamics in terms of $y_t \equiv Q^{-1}x_t$

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- Types of steady states:
 - 1 **sink**: if $|\lambda_1| < 1$ and $|\lambda_2| < 1 \Rightarrow x_t \rightarrow 0$
 - 2 **source**: if $|\lambda_1| > 1$ and $|\lambda_2| > 1 \Rightarrow x_t \rightarrow \infty$
 - 3 **saddle point**: if $|\lambda_1| > 1$ and $|\lambda_2| < 1$ (or vice versa)

Example

BLANCHARD-KAHN METHOD

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 - ① **state variables** predetermined at the beginning of t
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- Notation:
 - m is the number of control variables
 - n is the number of eigenvalues greater than one $|\lambda_i| > 1$
- **Blanchard-Kahn Principle:**
 - ① $n = m \Rightarrow$ unique solution
 - ② $n < m \Rightarrow$ multiple solutions
 - ③ $n > m \Rightarrow$ no solution

Illustration

Example

- Solve dynamic system with state x_{1t} and control x_{2t} :

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- Given $y_t = Q^{-1}x_t$, we get the **saddle path**

$$x_{2t} = -x_{1t}$$

LINEARIZATION

First-Order Approximations

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- In contrast, most macro models are *non-linear*
- The first-order approximations are used to bridge this gap:
 - ① **point of approximation**: steady state or initial equilibrium
 - ② **linearization**: deviations in levels or logs

Log-Linearization

- General approach:

$$X_t = \bar{X} \frac{X_t}{\bar{X}} = \bar{X} e^{\log \frac{X_t}{\bar{X}}} \equiv \bar{X} e^{x_t} \approx \bar{X}(1 + x_t)$$

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- Implications:

- power function

$$Z_t = X_t^a \quad \Rightarrow \quad z_t = ax_t$$

- multiplicative

$$Z_t = X_t Y_t \quad \Rightarrow \quad z_t = x_t + y_t$$

- additive

$$Z_t = X_t + Y_t \quad \Rightarrow \quad z_t = ax_t + (1 - a)y_t, \quad a \equiv \frac{\bar{X}}{\bar{Z}}$$

SAVINGS PROBLEM

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- ② **Log-linearization:** define $\alpha \equiv \frac{\bar{Y}}{\bar{B}}$ and obtain

$$x_{t+1} = Ax_t + Dy_t,$$

$$x_t = \begin{pmatrix} b_t \\ c_t \end{pmatrix}, \quad A = \begin{pmatrix} R & -(R - 1 + \alpha) \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad D = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}$$

Example

③ Diagonalize matrix A :

$$\Lambda = \begin{pmatrix} R & 0 \\ 0 & 1 \end{pmatrix}, \quad Q^{-1} = \begin{pmatrix} 1 & -(1 + \frac{\alpha}{R-1}) \\ 0 & \frac{1}{R-1} \end{pmatrix}, \quad Q^{-1}D = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}$$

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- ⑤ Find law of motion for b_t : substitute c_t into BC

$$b_{t+1} = b_t + \frac{\alpha}{R} y_t$$

CONTINUOUS TIME

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